Hyperspectral Intrinsic Image Decomposition Based on Physical Prior-Driven Unsupervised Learning

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***Abstract*— Deep learning-based intrinsic image decomposition (IID) has gained significant attention in computer vision due to the high efficiency and accuracy of learning-based methods. However, the development of deep learning-based IID methods in the remote sensing field has been limited by the lack of experimental datasets. This article proposes a two-stream encoder–decoder network for the single hyperspectral (HS) task. The proposed network comprises one reflectance estimation subnetwork (RES) and one shading estimation subnetwork (SES), which predict intrinsic properties separately. The proposed model introduces three physical losses to enhance performance: 1) in the RES, the self-similarity loss on the reflectance component is added to satisfy the basic assumption that pixels with similar intensity tend to have a similar reflectance property; 2) in the SES, the shading structure loss is added to ensure that the structure of the shading component conforms to physical observation; and 3) reconstruction loss connecting two subnetworks is required to ensure that the estimated intrinsic components are physically correct. Finally, to avoid an unreasonable composition, the entire network is initialized by reflectance estimated by the physical model. The quantitative experimental results of intraclass consistency and classification metrics demonstrate that the proposed physical prior-driven unsupervised learning-based IID network outperforms the current available learning or optimization-based approaches.**

***Index Terms*— Hyperspectral (HS), intrinsic image decomposition (IID), learning, reflectance, shading.**

I. Introduction

REMOTE sensing systems equipped with hyperspectral (HS) sensors can collect data by measuring the energy reflected from the Earth's surface at different wavelengths. HS images are known for providing abundant spectral information, which is beneficial for numerous applications, such as environmental monitoring [1], [2], target recognition [3], [4], change detection [5], [6], and classification [7], [8]. Hyperspectral intrinsic image decomposition (HyperIID) aims to recover the reflectance component from the original HS image and remove the shading component, where the former corresponds to the material-dependent properties of the ground surface and the latter corresponds to the interaction between environmental factors, such as illumination and surface orientation [9]. As the reflectance component plays a critical role in HS image processing tasks, HyperIID has gained much attention and has undergone significant development in recent years [10], [11].

Despite being studied for years, HyperIID remains a challenging open problem for two main reasons. First, the problem is highly ill-posed. Theoretically, there are countless factorizations of reflectance and shading that can reconstruct a given image, while finding the physically most probable answer is nontrivial. Second, acquiring ground truth remote sensing intrinsic images is difficult [12]. Therefore, most existing HyperIID methods are nonlearning-based models that solve the IID problem with reliable physical priors. Experimental results have demonstrated that this is an efficient approach to solve this challenging problem.

As an example, Kang et al. [13] proposed a new feature extraction method based on pixelwise intrinsic image decomposition (PIID) for HS image classification. They first reduced the spectral dimension of the HS image with image fusion and divided the dimension-reduced image into several subsets of adjacent bands. They then obtained the reflectance and shading components of each subset by using an optimization-based intrinsic image decomposition technique. Their experiments demonstrated that HyperIID is a powerful approach to improve classification performance. Furthermore, Jin and Gu [14] exploited the abundant spectral information in HS images and proposed a novel superpixel-based intrinsic image decomposition (SIID) framework for a single HS image. They built two superpixel segmentation maps of different scales and modeled the decomposition problem in a matrix form based on the finer superpixel map. They defined a constraint matrix by integrating the information in the coarser superpixel map. Experimental results showed its outstanding performance on HyperIID issues. To reduce the uncertainties and improve the recovery accuracy, Jin et al. [15] proposed another HyperIID method (HIID) with reflectance prior based on L1-graph coding, which can efficiently improve the spectral reflectance homogeneity within a class and preserve image boundaries.

In contrast, the problem of intrinsic image decomposition in the computer vision field has been studied for decades [16], [17], [18]. However, since the subject of study in computer vision is typically red, green, blue (RGB) images, there is inadequate spectral prior information provided, which limits the development of optimization-based methods. Therefore, learning-based methods have been proposed to improve the efficient of solving this challenging problem, for example,

Manuscript received 14 March 2023; revised 4 May 2023; accepted 21 May 2023. Date of publication 30 May 2023; date of current version 14 June 2023. This work was supported in part by the Distinguished Young Scholars of National Natural Science Foundation of China under Grant 62025107. *(Corresponding author: Yanfeng Gu.)*

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Digital Object Identifier 10.1109/TGRS.2023.3281490

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Barron and Malik [19] presented an IID method to recover shape, illumination, and reflectance components named shape, illumination, and reflectance from shading (SIRFS) based on learned properties of natural images. Moreover, deep learning-based methods include both supervised and unsupervised methods.

Due to the availability of large amount of ground truth intrinsic datasets [20], [21], performing IID tasks using supervised learning methods has become extremely practical. Initially, researchers trained their networks on synthetic datasets. For instance, Wang and Lu [22] presented a discriminative feature encoding network that extracts the features of different intrinsic layers, which is trained with the Max Planck Institute (MPI)-Sintel dataset [23]. Lettry et al. [24] designed a deep adversarial residual network with the MPI dataset as well. Shi et al. [25] trained an encoder–decoder convolutional neural network (CNN) with a rendered large-scale object intrinsic database based on 3-D models in the ShapeNet dataset. However, networks trained on synthetic datasets show low signal-to-noise ratio and nonrealistic sensor properties when used for real scenes. To improve the suitability of supervised networks, researchers have developed supervised learning-based IID for real scenes [26], [27], [28]. Zoran et al. [29] proposed a framework to infer midlevel vision by learning the ordinal relationship between pairs of points based on the intrinsic images in the wild (IIW) dataset, which is a human-labeled real-world dataset [30]. Nestmeyer and Gehler [31] also used the IIW dataset to train a CNN architecture with a filtering technique. These methods achieve impressive results in indoor scenes. However, the sparsity of annotations in the IIW dataset and the prevalence of indoor scenes limit the effectiveness of supervised learning-based IID networks on natural outdoor scenes, particularly on large-scale remote sensing images. The inherent differences between indoor small-scale images and natural remote sensing images cause a trained supervised learning model to lose its efficiency on the training dataset [32].

Obtaining ideal ground truth decompositions is difficult, and therefore, several studies have been conducted on solving the IID problem using unsupervised methods. Kinoshita and Kiya [33] introduced a self-supervised framework that considers reflectance consistency by using a color-illuminant model and training with losses from various illumination conditions. Some studies have developed unsupervised learning methods that rely on illumination-changing image sequences [34], [35]. Li and Snavely [36] designed a network that exploits information from image sequences and can successfully be applied to single views. Liu et al. [37] presented an unsupervised framework that learns latent features of reflectance and shading from unsupervised and uncorrelated data, including independence, domain-invariant content constraint, and physical constraint. Zhang et al. [38] further presented an unsupervised framework called unsupervised intrinsic decomposition network (UIDNet) to learn decomposition from a single image by training with the image itself and achieved good performance on real-world datasets. This framework is free from the restraints of a dataset, which is important in the remote sensing field.

In this article, a physical prior-driven unsupervised learning network inspired by UIDNet method is proposed to handle the HyperIID problem with a single HS image. It is a two-stream encoder–decoder network consisting of one reflectance estimation subnetwork (RES) and one shading estimation subnetwork (SES) with three physical loss functions: self-similarity loss, shading structure loss, and reconstruction loss. The input is a hybrid physical data that combines the original HS image and the output of a physical IID model, using the HIID model proposed by Jin as the physical IID model. Since the IID problem is severely ill-posed, theoretically, the same HS image can have countless valid decomposition results. Therefore, the network parameters are initialized by training with the output of the physical model to ensure that the decomposition is physically reasonable and to obtain better learned intrinsic components quantitatively.

This work makes the following contributions.

1) A novel unsupervised learning-based method is proposed to solve the HyperIID problem with a single HSI, utilizing physical loss functions. This approach breaks the limitations of optimization-based methods and enhances IID performance, providing a new approach to handle the problem.

2) The proposed method boosts the capability of HyperIID by integrating unsupervised learning networks with a physical prior-driven system. Experimental results demonstrate that it outperforms the state-of-the-art HyperIID methods.

The rest of this article is organized as follows. In Section II, we introduce the related works about IID. Section III describes the proposed approach in detail. The experimental results and analysis are shown in Section IV. Finally, the conclusion is drawn in Section V.

II. Related Work

In this section, the state-of-the-art physical model called HIID and common architecture used in IID network are introduced in detail.

A. Physical Model HIID

In remote sensing field, a single HS image can be decomposed into two intrinsic components, which are reflectance and shading components. It can be expressed in matrix form as follows:

$$\mathbf{I}(\lambda) = \mathbf{R}(\lambda) \circ \mathbf{S}(\lambda) \tag{1}$$

where $\mathbf{I}$ denotes the HS image, $\mathbf{R}$ denotes the reflectance component, $\mathbf{S}$ denotes the shading component, λ is the wavelength, and $\circ$ denotes the elementwise multiplication operator. With the help of HyperIID work, the shading component can be removed, and only the reflectance is retained. This greatly benefits HS applications, such as classification tasks and so on.

On the basis of decomposition model (1), Jin et al. further decomposed the shading into two components including environmental illumination and surface orientation to reduce the number of unknown variables. Additionally, they proposed a reflectance prior based on L1-graph coding to describe the relation between pigment density and reflectance. In this way,

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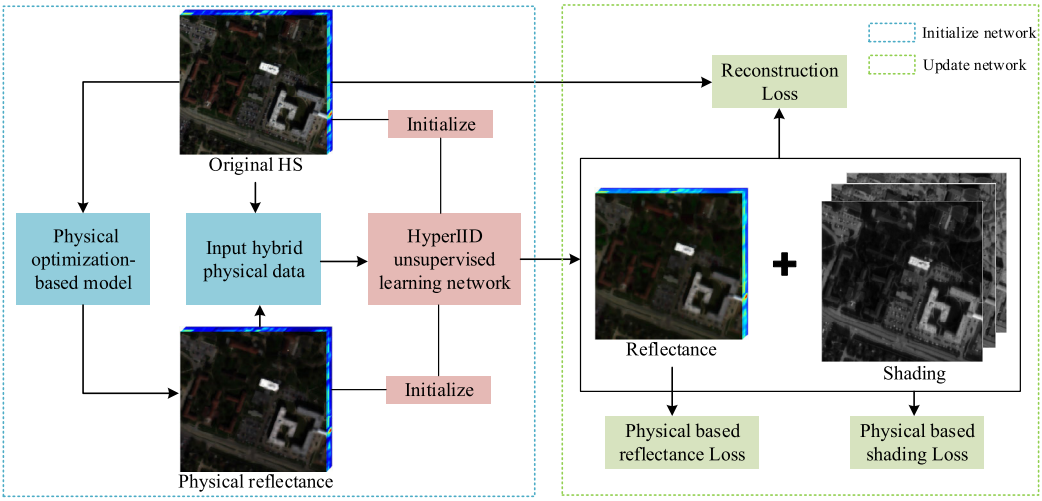


Fig. 1. Flowchart of the proposed method.

a noniterative optimization-based method was developed to recover a physical reflectance from a single HS image, which showed outstanding performance in HS image classification tasks [15]. The problem formulation can be described as

$$E(\rho, \beta, \mathbf{L}) = \left\|\rho + \mathbf{L}\mathbf{1}_n^T + \mathbf{1}_d\beta - \mathbf{O}\right\|_F^2 + \frac{1}{\sigma}\left\|\rho - \rho\mathbf{W}^T\right\|_F^2 \quad (2)$$

where $\boldsymbol{\rho}$ is a log-reflectance component, $\boldsymbol{\beta}$ is a log-orientation component, $\mathbf{L}$ is a log-illumination component, $\mathbf{1}_n$ and $\mathbf{1}_d$ denote the $n \times 1$ and $d \times 1$ vectors with all 1 s, respectively, and $\mathbf{O}$ is the input log-HS image.

In general, based on reasonable physical model, the HIID method has advantages in keeping spectral consistency and has good performance on classification tasks.

B. Encoder–Decoder Architecture in IID Network

Encoder–decoder architecture is a commonly employed approach in both supervised and unsupervised IID networks. Generally, the input of IID network is either a single image or an image sequence, while the output consists of two components: reflectance and shading. As a result, several practical architectures have been developed, such as one encoder and two decoders architecture, the two encoders and two decoders architecture, and the autoencoders architecture.

Li and Snavely [39] proposed a CNN-based model with one encoder and two decoders, incorporating skip connections, and trained it with ground truth data. Shu et al. [40] designed an end-to-end generative adversarial network with autoencoders to disentangle the latent content of intrinsic face properties. Chen and Koltun [16] introduced a two-stream IID network with two encoders and two decoders to predict albedo and shading separately. They employed three residual dilated blocks as decoders to reconstruct intrinsic properties from feature maps. Another encoder–decoder architecture with one shared encoder and three decoders was designed to predict the three intrinsic properties separately [15]. These studies demonstrate the practicality and reliability of the encoder–decoder architecture in solving IID problems.

In fact, combining residual network with encoder–decoder architecture not only appears in the case of IID networks. In other image processing tasks, such as image segmentation, the residual network has also been adopted into encoder–decoder architecture [41], [42], [43], [44], which has achieved great performance in experimental results.

On the one hand, HIID, an optimization-based method designed based on physical priors, has shown outstanding performance on reflectance recovery tasks. However, it is always faced with the problem of manually tuning parameters of constraints and low efficiency. On the other hand, although existing unsupervised learning-based IID networks can be directly utilized for HS image, the training pattern designed for RGB images is not suitable for remote sensing images. This is because the input HS image is not white-balanced as assumed for RGB images and the shading component is not single channel since the shading component in HS images is a product affected by illumination. To address these problems, a physical prior-driven unsupervised learning network is proposed by constructing a two-stream encoder–decoder network with three physical loss functions and a hybrid physical data as an input. This approach boosts the performance of HyperIID and improves the decomposition efficiency for HS images.

III. Proposed Method

Fig. 1 shows the overall flowchart diagram of the proposed method, which comprises two steps: first, network parameters will be initialized driven by physical prior in initialization step and then in updating step, network parameters will be updated with observed loss functions.

A. Physical Prior-Driven Model

According to (1), the fundamental purpose of HyperIID is to decompose input HS image $\mathbf{I}$ into the pixelwise product of reflectance component $\mathbf{R}$ and shading component $\mathbf{S}$. Clearly, the HyperIID problem is an ill-posed and challenging problem as it is underconstrained, and there are more unknown variables than knowns. To promote the separability of two intrinsic

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properties, three loss functions based on physical observation are introduced into the HyperIID network. Two of these loss functions constrain RES and SES, respectively, while the third ensures that the two output components can be reconstructed as the original HS image. The energy optimization formulation can be expressed as follows:

$$\min_{\mathbf{R},\mathbf{S}} \ \Phi(\mathbf{R}) + \Gamma(\mathbf{S}) \quad \text{s.t.} \ \ \mathbf{I} = \mathbf{R} \circ \mathbf{S} \tag{3}$$

where $\Phi(\cdot)$ denotes a constraint on RES, and $\Gamma(\cdot)$ denotes a constraint on SES, both will be discussed in detail later.

In most cases, unsupervised IID networks for RGB images are able to achieve good decomposition results under constraint presented in (3). However, this is not the case for HS images. Specifically, with the decomposition model in (1), each pixel of an input RGB image refers to only three equations with four unknowns, whereas an HS image has equations with spectral dimension quantities and double unknowns. Therefore, the undetermined IID system for HS images is more severely ill-posed than that for RGB images. To avoid the parameters of HyperIID network optimizing to an erroneous decomposition result, as shown in Fig. 1, the proposed HyperIID network has a two-stage training process. The first stage is a physical prior-driven stage to initialize the parameters of HyperIID network with original HS image and output image of optimization-based IID method.

Moreover, the standard approach is to train the HyperIID network with the original HS image as input data, which can be used to estimate the optimized intrinsic properties. Alternatively, although the physical output of optimization-based IID model provides incomplete intrinsic properties due to simplified physical model of optimization-based IID methods, it can also be used for training since there is physical relation with the original HS image that cannot be learned by any IID network. Therefore, as shown in Fig. 1, the first step of physical prior-driven model is to generate physical output data and combine it with original HS image to improve the efficiency of HyperIID network. The simplest way to combine the two data is to concatenate them along spectral dimension axis into a hybrid physical data $\mathbf{D}$, which can be expressed as

$$\mathbf{D} = \left[\mathbf{D}_{\text{ori}}, \mathbf{D}_{\text{phy}}\right] \tag{4}$$

where $\mathbf{D}_{\text{ori}}$ denotes the original input HS image, and $\mathbf{D}_{\text{phy}}$ denotes the physical output image of optimization-based IID method.

B. Loss Functions

As shown in Fig. 1, the second stage of the proposed HyperIID network involves updating the network parameters using three loss functions based on physical observations: reconstruction loss, reflectance local similarity loss, and shading structure loss. Formally, these three loss functions are combined through linear addition, which can be expressed as follows:

$$\min_{\mathbf{R},\mathbf{S}} \ L_{\text{rec}}(\mathbf{R},\mathbf{S}) + \lambda_1 L_{\text{ref}}(\mathbf{R}) + \lambda_2 L_{\text{sha}}(\mathbf{S}) \tag{5}$$

where the reconstruction loss L_{rec} ensures the estimated reflectance and shading can reconstruct the input HS image, the reflectance local similarity loss L_{ref} aims to obtain a reflectance that has reasonable similarity relation of materials in a local window, and the shading structure loss L_{sha} is added to introduce shading structure information from input HS image. λ_1 and λ_2 are the weight parameters, which are set as 0.9 and 0.5 in the following experiments, respectively. These physical observation-based loss functions will be described one by one as follows.

1) Reconstruction Loss: According to the basic physical model of HS imaging represented in (1), the predicted reflectance $\mathbf{R}$ and shading $\mathbf{S}$ are supposed to reproduce the original HS image $\mathbf{I}$. Since input image is the pixelwise product of reflectance and shading, transforming (1) into log domain will convert multiplication operation to addition operation, which is more convenient and stable for network training and loss function calculation. Therefore, the reconstruction loss in log domain can be defined with L1 norm-based error metric as follows:

$$L_{\text{rec}}(\mathbf{R},\mathbf{S}) = \| \log \mathbf{R} + \log \mathbf{S} - \log \mathbf{I} \|_1 . \tag{6}$$

2) Local Similarity Loss for Reflectance: Based on the physical assumption that pixels with similar intensity in a local area are supposed to have similar reflectance spectrum, the similarity of local reflectance values can be measured using the spectral angle mapper (SAM) metric between neighboring pixels in an original input image. It should be noted that the quality of HS remote sensing imaging is often affected by many factors, such as illumination conditions, and there is almost never a spatially uniform distribution of illumination in such large-scale imaging scenes. However, the same material under different illumination energies will still have a similar spectrum in normalized spectral domain. Therefore, the local similarity loss for reflectance can be defined as follows:

$$L_{\text{ref}}(\mathbf{R}) = \left\| e^{-|\mathbf{A}[N(\mathbf{I})]|} \circ |\nabla_2 \mathbf{R}| \right\|_2^2 \tag{7}$$

$$\mathbf{A}(\mathbf{I}) = \left[\text{SAM}(\mathbf{I}_{i-1}, \mathbf{I}_i), \text{SAM}\left(\mathbf{I}_{j-1}, \mathbf{I}_j\right)\right] \tag{8}$$

$$\text{SAM}\left(\mathbf{I}_{j-1}, \mathbf{I}_i\right) = \arccos\left[\left(\mathbf{I}_{j-1}{}^T \mathbf{I}_j\right) / \left(\sqrt{\mathbf{I}_{j-1}{}^T \mathbf{I}_{j-1}} \sqrt{\mathbf{I}_j{}^T \mathbf{I}_j}\right)\right] \tag{9}$$

where $N(\cdot)$ is the normalization operation for input image, $\mathbf{A}(\mathbf{I})$ computes the spectral angle value of adjacent pixels along two spatial axes, and the gradient operator $\nabla_2 = [\nabla_{2x}, \nabla_{2y}]$ can be calculated by the forward difference between adjacent pixels.

3) Shading Structure Loss: Regarding RGB image decomposition, it is commonly accepted that the input RGB image is typically white-balanced, and the shading component only corresponds to the surface distribution of objects rather than illumination condition, so shading image of RGB image is single channel, and changes in shading only lead to proportional changes in channel bands. However, this assumption does not hold for HS images, as illumination is not spatially uniformly distributed in large-scale remote sensing scenes, making it difficult to maintain a white-balanced condition. Additionally, the shading component of HS images consists of

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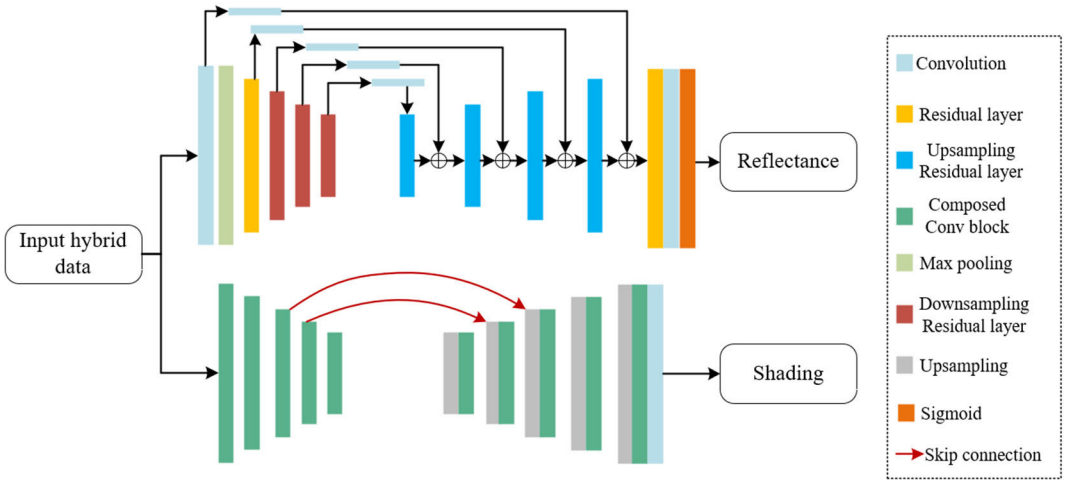


Fig. 2. Hyperspectral image intrinsic decomposition network (HyperIIDNet).

a spatially varying illumination component, which is not taken into account in optimization-based methods. Therefore, the shading estimation loss function is introduced with two parts: the first part considers the continuous areas of original HS image, where shading component is assumed to be continuous as well; and the second part considers discontinuous areas with similar reflectance, where changes can only be caused by shading. We establish the shading structure loss function to prevent large changes of shading in continuous areas and small changes of shading in discontinuous areas with similar reflectance components, which can be defined as follows:

$$L_{\text{sha}}(\mathbf{S}) = \left\| e^{-|\nabla_2 \mathbf{I}|} \circ |\nabla_2 \mathbf{S}| + |\nabla_2 \mathbf{I}| \circ e^{-\{|\mathbf{A}[N(\mathbf{I})]| + |\nabla_2 \mathbf{S}|\}} \right\|_2^2. \quad (10)$$

C. HyperIID Network Architecture

Fig. 2 illustrates the overall architecture of the proposed HyperIID network, which is composed of a two-stream encoder–decoder network with two distinct subnetworks: the RES and the SES. RES is a residual encoder–decoder network, whereas SES is a fully convolutional encoder–decoder network. In the context of HyperIID, the reflectance contains abundant spectral information pertaining to ground materials. As a result, the network is faced with the challenge of accurately recovering this information. To mitigate the issue of vanishing gradients, residual unit with short paths between input and output is applied to boost the reflectance recovery ability of network.

The RES employs Resnet-50 architecture as a residual unit [45], with the average pooling and fully connected layers removed. In the encoder stage, the first downsampling operation is carried out using a 7×7 convolution layer with stride 2, followed by a batch normalization layer and rectified linear unit (ReLU) function [46]. The second downsampling layer is a 3×3 maxpooling layer with stride 2. After these two downsampling operations, there are four residual layers comprising of three, four, six, and three residual units, respectively. The first residual layer does not have downsampling ability, whereas the others have a downsample residual unit to downsample feature maps and increase feature channels with a stride of 2. In decoder stage, there are five residual layers, with the first four of them using upsample residual units to achieve the upsampling purpose. The upsample unit described in [31] is employed, and the last residual layer is followed by a 2×2 convolution transpose layer. Specifically, the encoder stage uses the standard bottleneck, while the decoder stage uses a standard residual building block. At the end of RES, a sigmoid activation function is applied to normalize the output reflectance to the range [0, 1]. Furthermore, a connection from encoder to decoder is employed by adding a 1×1 convolution layer, followed by a batch normalization layer and ReLU function to transfer feature maps of different channel sizes to lower sizes. The SES architecture includes five convolution blocks in encoder stage, where each block consists of two convolution blocks. The first block comprises a convolution layer with stride 2, followed by a batch normalization layer and ReLU function. The second convolution layer has a stride of 1. In the decoder stage, all convolution blocks have a stride of 1, and bilinear upsampling is applied to each decoder layer to increase feature maps while decreasing feature channels. Similarly, skip connections are also used in SES, connecting only the third and fourth encoder layers with their corresponding decoder layers, as deeper layers contain most of the shading structure information. Finally, a 1×1 convolution layer is used to output shading estimation at the end of SES.

D. Initialization and Setting Details

The dimensionality setting for each layer of the proposed HyperIID network structure is listed in Table I. It is optimized using the Adam optimizer with a learning rate of 10^{-3}, and the total number of iterations is set to 10 000, which is sufficient for HyperIID network to converge to a good result. Additionally, as discussed in Section III-A, in order to avoid the parameters of HyperIID network optimize to an erroneous decomposition result as the IID problem for HS image is severely ill-posed, it is necessary to initialize the parameters of HyperIID network with original HS image and output image of optimization-based IID method. This initialization process uses both reconstruction loss and initialization loss

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TABLE I
DIMENSIONALITY SETTING OF LAYERS

RES Subset			SES Subset		
Block	Input	Outpu	Block	Input	Output
Conv1	96	128	Conv1	96	128
Residual Layer1	128	256	Conv2	128	256
Residual Layer2	256	512	Conv3	256	512
Residual Layer3	512	1024	Conv4	512	1024
Residual Layer4	1024	2048	Conv5	1024	2048
Residual Layer5	512	256	Conv6	2048	2048
Residual Layer6	256	128	Conv7	2176	1024
Residual Layer7	128	128	Conv8	1280	512
Residual Layer8	128	64	Conv9	512	256
Residual Layer9	64	64	Conv10	256	128

for 1000 iterations. The initialization loss can also be defined using an L1 norm-based error metric

$$\min L_{\text{rec}}(\mathbf{R}, \mathbf{S}) + L_{\text{ini}}(\mathbf{R}) \tag{11}$$

$$L_{\text{ini}}(\mathbf{R}) = \left\|\mathbf{R} - \mathbf{D}_{\text{phy}}\right\|_1. \tag{12}$$

The HyperIID network performs better after undergoing physical prior-driven stage processing and being trained with three other physical observation-based losses. In this article, the HIID method was utilized to generate an estimated physical reflectance result for training the HyperIID network. The parameters in HIID method are set according to the author's suggestions. Thus, by utilizing an unsupervised learning network model with a proposed training strategy, the HyperIID network has great potential for recovering optimal reflectance component results.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, some experimental results are reported to illustrate the effectiveness of the proposed HyperIID unsupervised learning network with physical prior-driven strategy.

A. Data Description and Experimental Setting

The experiments are validated on four datasets including two HS images obtained by ZY1-02D satellite, one HS data provided by the 2018 IEEE GRSS Data Fusion Contest and one pseudo synthetic HS image produced from it: 1) the Yellow River HS image; 2) the Yellow River-area2 HS image; 3) real Houston HS image; and 4) synthetic Houston HS image. The ground truth maps of the first two datasets are manually produced, while the ground truth map of Houston data was provided from Fusion Contest.

1) Yellow River-Area1: This HS image was acquired by the ZY1-02D sensor over Yellow River Wetland Delta in China at June 28, 2020. ZY1-02D sensor operates in the wavelength range from 0.4 to 2.5 mm, and its spatial resolution is about 30 m. The size of the image is 192 × 192 pixels, and the image originally has 156 spectral bands, Fig. 3(a) shows the false color images of this dataset, and Yellow River-area1 data cover farmland and river of Yellow River Delta.

2) Yellow River-Area2: This HS image was another area acquired by the ZY1-02D sensor over Yellow River Wetland Delta with 192 × 192 pixels at July 24, 2020. Its data parameters are as the same with Yellow River-area1 HS image, and we use all the 156 bands to perform subsequent experiments. Fig. 3(b) shows the false color images of this dataset, and we can see it mainly covers area of salt ponds and fish culture zones.

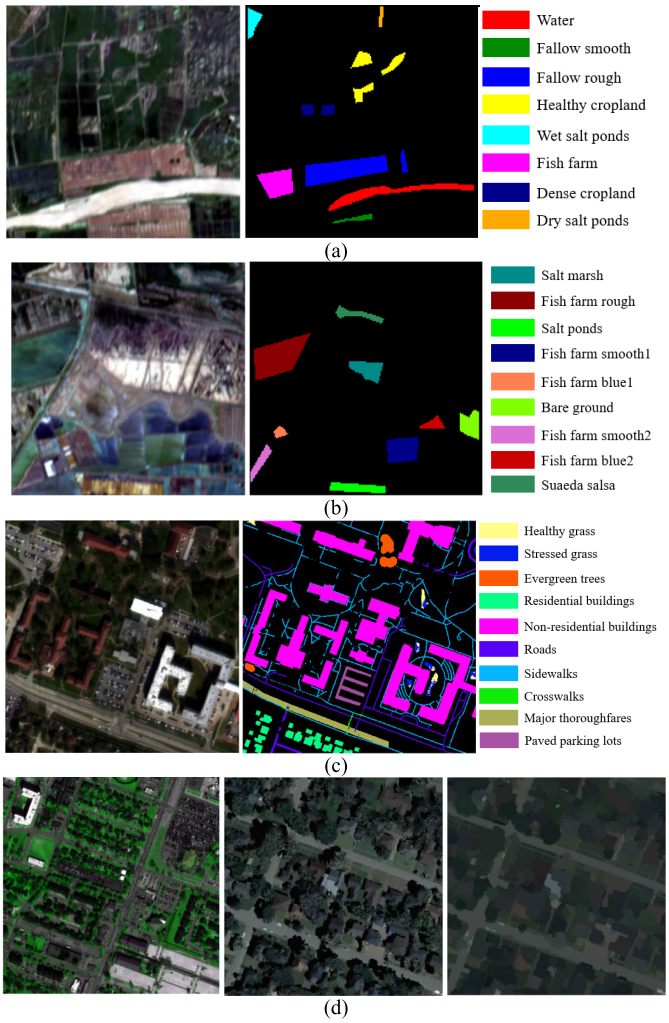


Fig. 3. False color images and ground reference maps of four datasets. (a) Yellow River-area1. (b) Yellow River-area2. (c) Real Houston subscene. (d) Synthetic Houston subscene.

3) Real Houston Subscene: The HS image was acquired by the National Center for Airborne Laser Mapping (NCALM) using ITRES CASI 1500 hyperspectral camera on February 16, 2017, which covers a 380–1050-nm spectral range with 48 bands at a 1-m ground sampling distance (GSD). They also provide ground truth data corresponding to 20 urban land use and land cover classes at a 0.5-m GSD. For the convenience of experiments, we cropped an area with size of 416 × 416 pixels. Fig. 3(c) shows the false color images of this dataset. These cropped data are mainly covered with buildings and urban street trees.

4) Synthetic Houston Subscene: A pseudo synthetic HS image with ground truth reflectance data is used to verify the accuracy of reflectance components estimated by the proposed method. These data are a subscene with a size of 200 × 200 pixels cropped from Houston dataset provided by 2018 IEEE GRSS Data Fusion Contest and are produced by research [12] with several automatic and manual processing steps.

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TABLE II
WCRV RESULTS OF DIFFERENT METHODS FOR THREE DATASETS

No.	Yellow River-area1				Yellow River-area2				Real Houston			
	Original	UIDNet	HIID	Proposed	Original	UIDNet	HIID	Proposed	Original	UIDNet	HIID	Proposed
C1	21.4	10.9	**4.5**	4.6	16.2	**2.9**	7.4	6.5	10.9	**1.3**	2.8	2.7
C2	2.0	0.2	**0.1**	0.2	11.3	4.6	2.5	**2.3**	72.3	40.1	16.1	**15.0**
C3	85.5	47.8	18.7	**17.9**	32.4	**7.9**	9.3	11.0	39.5	12.5	7.0	**6. 6**
C4	32.8	**2.4**	6.5	5.6	30.8	61.1	**7.7**	7.8	131.9	114.8	21.9	**18.8**
C5	28.7	39.0	4.0	**1.9**	2.4	1.8	**0.6**	0.9	25970.4	36318.7	7657.7	**7348.2**
C6	47.6	15.3	6.7	**6.4**	16.1	9.2	4.3	**2.7**	439.0	382.5	98.0	**92.9**
C7	4.4	**0.5**	0.9	4.1	7.2	11.0	3.0	**2.0**	1745.1	1295.4	346.9	**318.3**
C8	8.1	2.7	0.8	**0.3**	1.4	0.4	0.4	**0.3**	6.3	2.4	**1.1**	**1.1**
C9					3.0	2.6	**0.7**	**0.7**	84.1	38.8	12.6	**11.5**
C10									20.9	6.3	3.9	**3.6**
Total	230.5	118.8	42.2	**41.0**	120.8	101.5	35.9	**34.2**	28520.4	38212.8	8168	**7812.1**

Since reflectance component is supposed to have better spectral homogeneity within a class, therefore, we utilize within-class reflectance variance (WCRV) metric to measure the quality of estimated intrinsic reflectance image, which can be defined as

$$\mathrm{WCRV}(\mathbf{R}) = \sum_{i=1}^{k} \sum_{j \in C_i} \left\| \mathbf{R}_j - \mathbf{m}_i \right\|_2^2 \tag{13}$$

where k is the total number of categories, and $\mathbf{m}_i$ is the mean vector of category C_i in reflectance.

In addition, with synthetic HS data, two metrics, such as mean squared error (MSE) and local MSE (LMSE), are used to evaluate the IID ability of different methods (lower is better) [16].

Besides, intrinsic reflectance component is supposed to be helpful for hyperspectral classification tasks, two common metrics are used in our study to evaluate the quality of the classification accuracy, and they are overall accuracy and Kappa coefficient.

1) *Overall accuracy (OA):* It is a useful metric to evaluate the effectiveness of classification performance, which measures the proportion of correctly classified samples in a dataset.

2) *Kappa coefficient (Kappa):* It is used to evaluate the reliability that takes into account the agreement that can be expected by chance.

The standard KNN classifier is employed to perform the classification task for intrinsic reflectance images produced by comparison methods. There are three different settings for the number of training samples: 1%, 5%, and 10% training samples per class for all datasets. In addition, all reported classification accuracy results are the average and standard deviation of ten experiments on random training and test samples.

Different comparison methods are described as follows.

1) Raw: original input hyperspectral image.

2) SIRFS: a supervised intrinsic decomposition method that can recover reflectance from a masked object.

3) UIDNet: an unsupervised intrinsic decomposition network designed for natural RGB images.

4) HIID: a physical optimization-based hyperspectral intrinsic decomposition method with two intrinsic priors.

5) SIID: an optimization-based hyperspectral intrinsic decomposition method using superpixel technology.

6) Adaptive filtering based intrinsic image decomposition (AFIID): a supervised learning-based intrinsic image decomposition method applying adaptive filtering technique to CNN learning of reflectance prediction [31].

7) HyperIIDNet-stage2: reflectance is produced by the proposed HyperIID network, which is only trained with aforementioned three physical losses without physical prior-driven stage.

7) Proposed: reflectance result is estimated by using HyperIID network, which is driven by physical prior knowledge.

The parameters of compared methods are tuned individually according to the authors' suggestions. Additionally, the scale size of subpixels in the SIID method is appropriately adjusted to match the input image size.

B. Experimental Results on Real Datasets

1) Analysis of Reflectance Recovery Quality: The WCRV metric assesses the homogeneity of reflectance performance for a given ground surface category, and a lower score indicates better performance. Table II presents the total WCRV metric results for each category, as well as the overall scores, which are computed using labeled samples from three real datasets. The results indicate that the proposed method significantly improves within-class reflectance consistency by removing the influence of environmental interference factors. Specifically, it achieves an average increase of approximately 70% for the total WCRV score on three datasets compared to the original HS image. Furthermore, the proposed method outperforms both learning-based and optimization-based methods in terms of WCRV performance, achieving higher total scores and better performance across a greater number of categories on three real datasets. It exhibits an average increase of approximately 65% and 4% for learning-based and optimization-based methods, respectively, which highlights its efficiency in enhancing within-class reflectance homogeneity.

Figs. 4–6 show the visual results obtained from different comparison methods. The results indicate that the learning-based method, SIRFS, suffers from a spatial distortion with severe vertical stripe noise, which is caused by wrong estimation of depth information. The unsupervised learning-based

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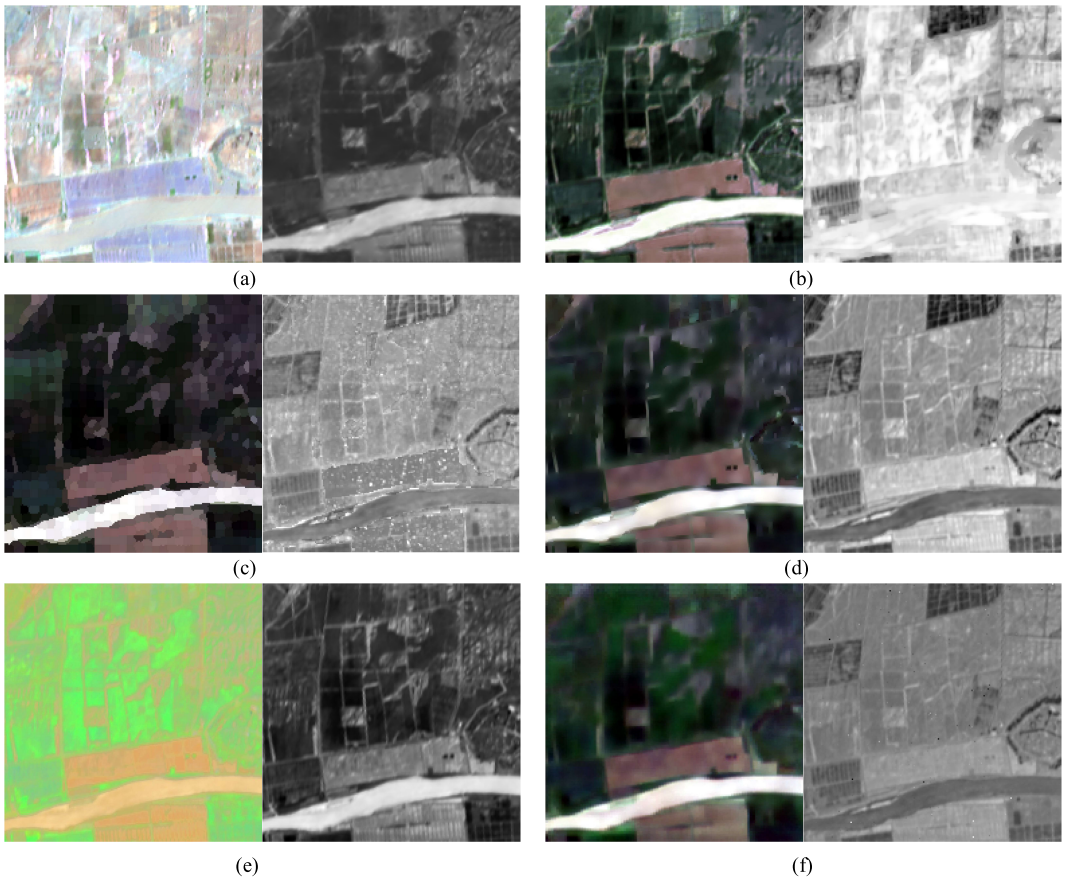


Fig. 4. Decomposition results comparison for Yellow River-area1 dataset. The left one is decomposed reflectance and the right one is decomposed shading. (a) SIRFS. (b) UIDNet. (c) SIID. (d) HIID. (e) AFIID. (f) Proposed.

method UIDNet does not consider the actual situation of hundreds of spectral channels, and its lack of a physical prior knowledge results in an erroneous decomposition, as evidenced by the abnormal white area in the top-right corner of the real Houston data. In contrast, the HIID method, specifically designed for HS images, effectively avoids spectral distortion problems and achieves good within-class reflectance homogeneity while preserving image boundaries. For the superpixel-based method SIID, it is obvious to find that it has severe traces of superpixel segmentation of composed reflectance images. For the supervised AFIID method, it is obvious to find that the decomposed reflectance images are faced with a severe spectral distortion problem and it is hard to tell ground objects by human eyes. Comparing the HyperIID method without a physical prior-driven system and the proposed method, it is evident that the physical prior-driven system greatly improves the HyperIID network's IID performance. The proposed network achieves better within-class spectral consistency and clearer edges than the HyperIID method.

2) Classification Performances Comparison: Reflectance is solely dependent on the properties of ground surface materials, making it a valuable aid for HS image classification. The classification results of comparison methods on three datasets are presented in Figs. 7–9. It is observed that the original HS image and HyperIID lacking a physical prior-driven system tend to produce noisy classification maps. Furthermore, the latter method misclassifies some materials since its network cannot accurately estimate the reflectance component without the aid of a physical prior boosting strategy. While the UIDNet method may seem to perform well in terms of spectral homogeneity within material categories, it actually exhibits more misclassification issues due to the spectral distortion problem inherent to the method. In addition, since SIRFS method tends to produce a reflectance component disturbed by vertical stripes noise, it is reflected on classification maps. For the SIID method, its classification result is still affected by segmentation blocks. Moreover, for the AFIID method, due to the severe spectral distortion problem, its classification ability is not even as good as the input HS image with low spatial resolution, such as Yellow River datasets, and the proposed method demonstrates the most stable and reasonable classification performance across all three datasets.

The numerical overall accuracy metric results of three real datasets are illustrated in Tables III–V. As shown in the table, the SIID method can get best classification result in Yellow River-area2 and real Houston data with certain training sample numbers, which is caused by different superpixel scale sizes and is not a stable performance for classification task. In contrast, the proposed method is superior to the state-of-the-art learning-based and optimization-based methods. For instance, in the case of the Houston dataset, the proposed method achieves an average 4.5% improvement over the original HS image. Compared to learning-based and optimization-based

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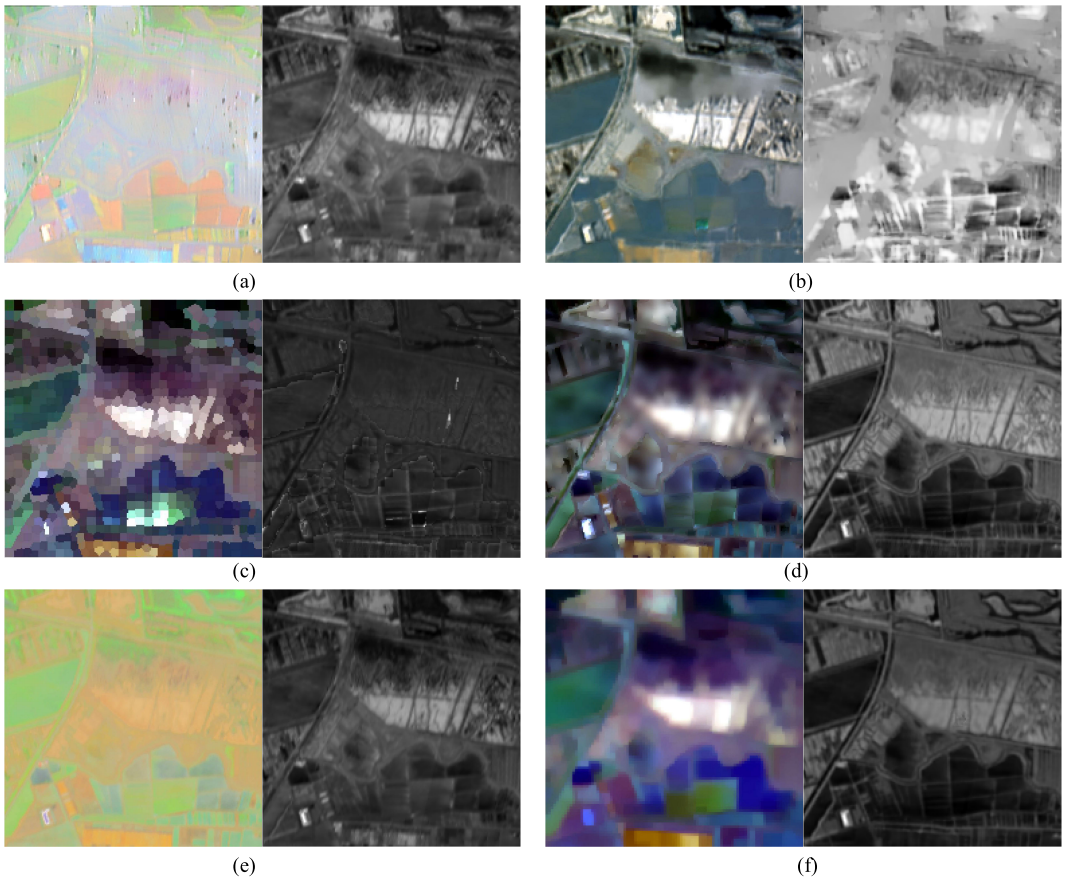


Fig. 5. Decomposition results comparison for Yellow River-area2 dataset. The left one is decomposed reflectance and the right one is decomposed shading. (a) SIRFS. (b) UIDNet. (c) SIID. (d) HIID. (e) AFIID. (f) Proposed.

TABLE III
CLASSIFICATION RESULTS OF DIFFERENT METHODS WITH DIFFERENT TRAINING SAMPLES FOR YELLOW RIVER-AREA1 DATASET

Methods	1%		5%		10%	
	OA (%)	Kappa (%)	OA (%)	Kappa (%)	OA (%)	Kappa (%)
Ori-HSI	80.09±5.99	74.08±7.82	95.17±0.89	93.81±1.15	96.77±0.59	95.86±0.76
SIRFS	61.10±9.38	48.13±11.97	91.38±2.39	88.88±3.16	97.17±0.77	96.38±0.99
UIDNet	86.38±5.35	82.20±6.97	97.32±1.21	96.56±1.58	98.81±0.63	98.49±0.80
HIID	83.86±6.17	79.33±7.73	97.10±0.71	96.27±0.91	98.56±0.53	98.16±0.68
SIID	82.32±7.94	76.99±10.44	96.80±1.26	95.90±1.61	98.74±0.58	98.39±0.74
AFIID	76.32±8.85	69.16±11.46	93.98±1.22	92.25±1.56	94.41±0.56	92.83±0.70
HyperIIDNet-stage2	80.71±7.27	74.72±9.77	97.07±0.51	96.24±0.66	99.32±0.71	99.14±0.91
Proposed	**87.04±4.51**	**83.18±5.88**	**98.52±1.29**	**98.10±1.67**	**99.56±0.47**	**99.44±0.60**

TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS WITH DIFFERENT TRAINING SAMPLES FOR YELLOW RIVER-AREA2 DATA

Methods	1%		5%		10%	
	OA (%)	Kappa (%)	OA (%)	Kappa (%)	OA (%)	Kappa (%)
Ori-HSI	66.02±4.41	55.78±5.86	94.88±1.81	93.67±2.26	97.68±0.80	97.15±0.99
SIRFS	60.42±4.40	49.88±6.31	94.24±0.97	92.92±1.19	96.46±0.78	95.66±0.95
UIDNet	66.70±6.11	56.90±8.03	96.12±1.18	95.21±1.44	98.28±0.32	97.90±0.39
HIID	68.05±8.00	58.59±11.05	96.16±1.92	95.30±2.41	98.94±0.59	98.70±0.73
SIID	**71.05±4.62**	**62.73±6.12**	95.77±1.48	94.80±1.84	97.66±1.47	97.12±1.81
AFIID	66.35±10.64	56.39±14.43	95.19±1.45	94.10±1.78	97.62±0.41	97.08±0.50
HyperIIDNet-stage2	70.65±8.43	61.89±11.38	97.59±0.97	97.04±1.20	99.75±0.45	99.69±0.55
Proposed	70.67±7.53	62.33±10.09	**98.00±1.89**	**97.55±2.32**	**99.82±0.24**	**99.78±0.29**

methods, it shows an average improvement of approximate 2% on OA performance. Furthermore, the experimental results support the hypothesis that a physical prior-driven strategy is highly beneficial, as demonstrated by the overall accuracy.

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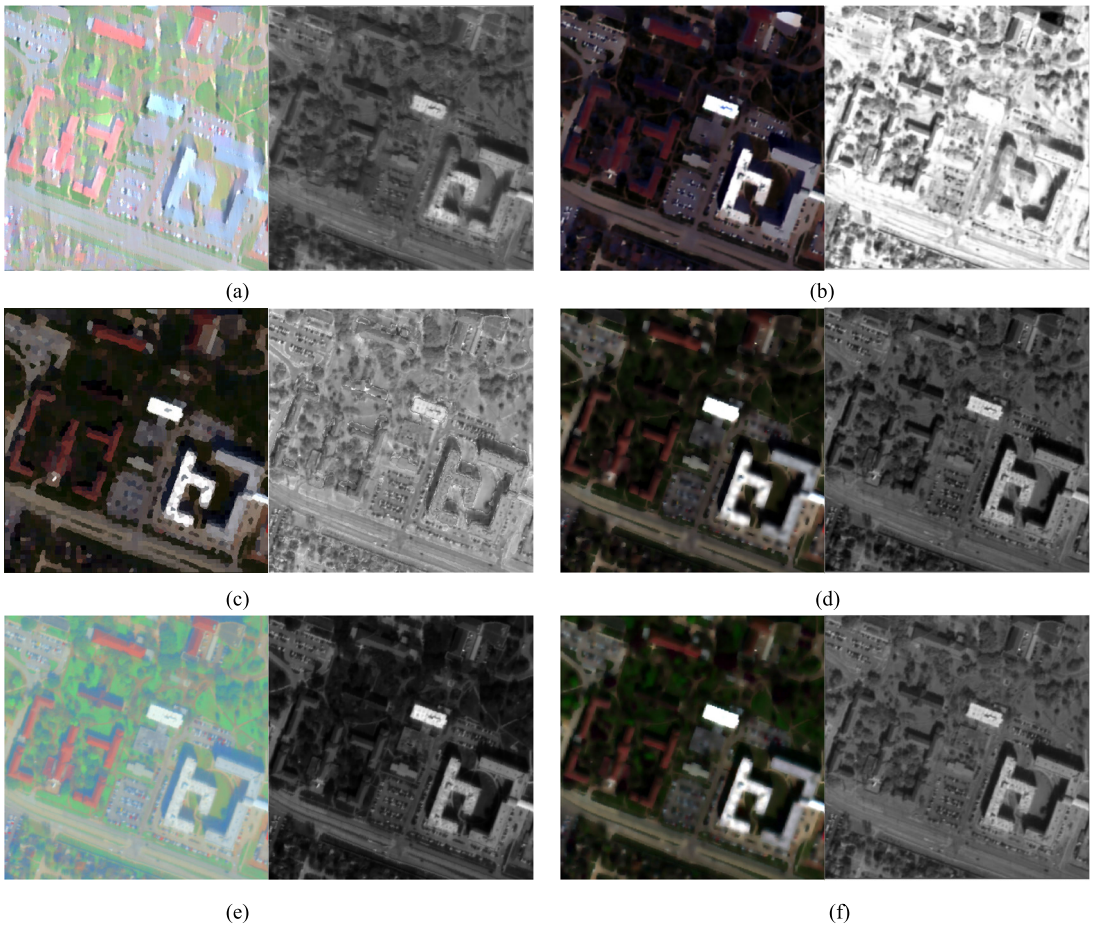


Fig. 6. Decomposition results comparison for Houston dataset. The left one is decomposed reflectance and the right one is decomposed shading. (a) SIRFS. (b) UIDNet. (c) SIID. (d) HIID. (e) AFIID. (f) Proposed.

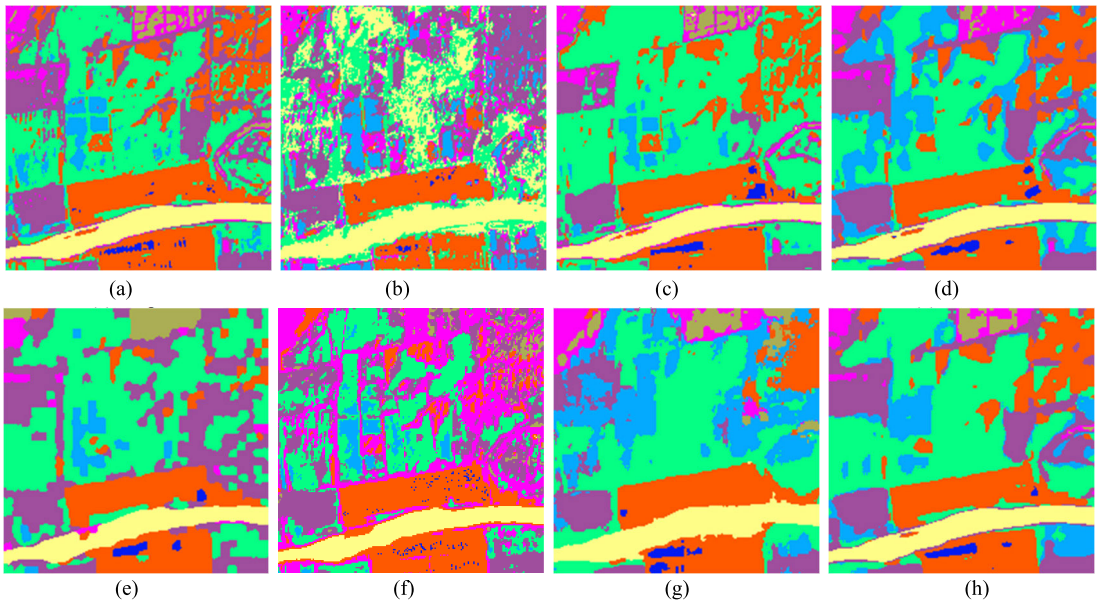


Fig. 7. Classification results of Yellow River-area1 data. (a) Original. (b) SIRFS. (c) UIDNet. (d) HIID. (e) SIID. (f) AFIID. (g) HyperIID-stage2. (h) Proposed.

In general, experimental results demonstrate that the proposed method surpasses current HyperIID methods in estimating hyperspectral reflectance components.

3) Analysis of Weights in Loss Function: There are two weights referring two loss functions in the proposed network, and they are weight of reflectance loss λ_1 and weight of shading loss λ_2. The influence of weight ratio changing is shown in Table VI. It can be seen that when the weight ratio is 2, the proposed network can obtain a relatively fine decomposition result.

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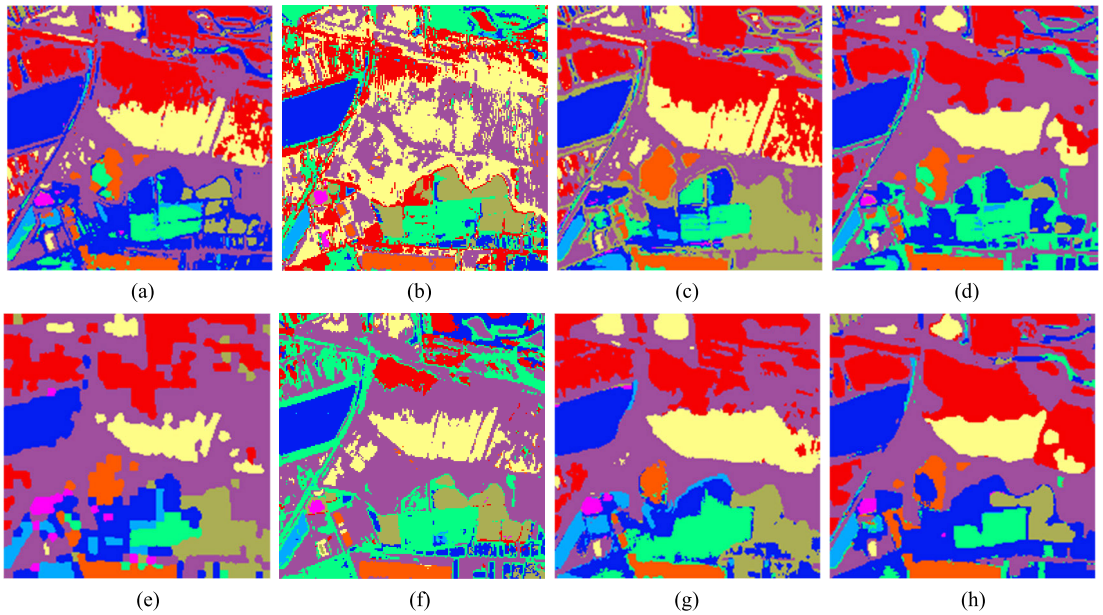


Fig. 8. Classification results of Yellow River-area2 data. (a) Original. (b) SIRFS. (c) UIDNet. (d) HIID. (e) SIID. (f) AFIID. (g) HyperIID-stage2. (h) Proposed.

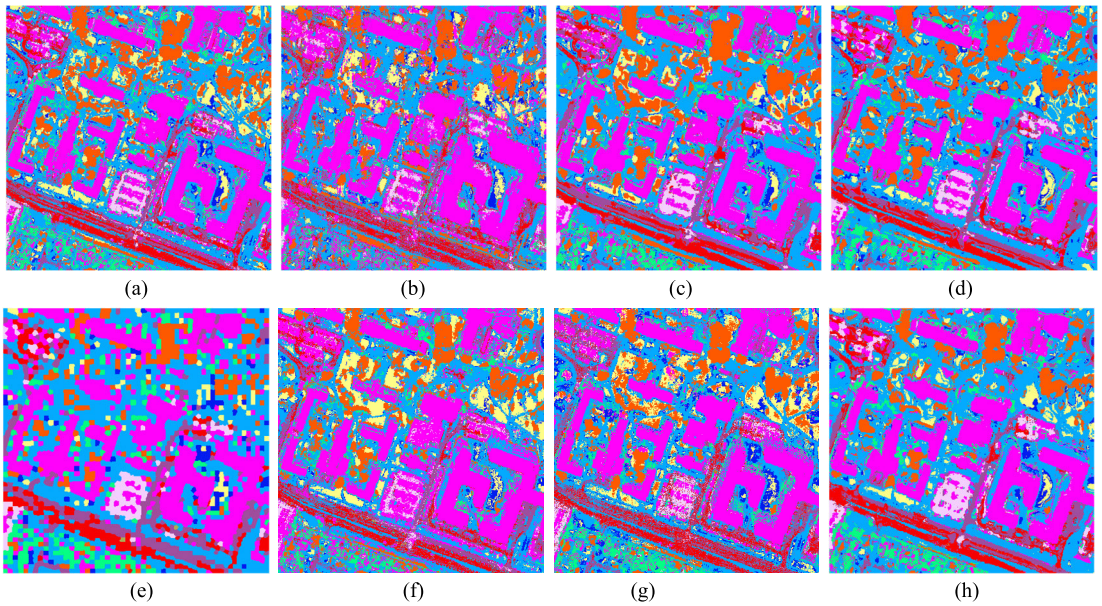


Fig. 9. Classification results of real Houston data. (a) Original. (b) SIRFS. (c) UIDNet. (d) HIID. (e) SIID. (f) AFIID. (g) HyperIID-stage2. (h) Proposed.

TABLE V
CLASSIFICATION RESULTS OF DIFFERENT METHODS WITH DIFFERENT TRAINING SAMPLES FOR HOUSTON DATASET

Methods	1%		5%		10%	
	OA (%)	Kappa (%)	OA (%)	Kappa (%)	OA (%)	Kappa (%)
Ori-HSI	73.12±0.81	57.36±2.24	79.83±0.32	68.85±0.40	82.06±0.21	72.31±0.28
SIRFS	63.69±0.96	41.61±0.92	76.45±0.35	62.74±0.52	81.41±0.35	70.78±0.53
UIDNet	76.36±0.75	63.33±1.22	81.65±0.22	71.71±0.34	83.35±0.15	74.40±0.23
HIID	74.56±0.96	60.10±1.38	81.96±0.35	72.15±0.55	84.34±0.25	75.87±0.35
SIID	73.57±0.68	58.52±1.23	82.13±0.40	72.53±0.58	**86.51±0.45**	**79.08±0.66**
AFIID	75.84±0.82	60.46±0.93	81.59±0.29	71.40±0.46	83.15±0.16	73.77±0.26
HyperIIDNet-stage2	75.65±0.54	62.41±0.66	82.43±0.21	73.07±0.34	85.10±0.14	77.08±0.21
Proposed	**76.48±0.65**	**63.35±0.94**	**83.46±0.27**	**74.44±0.44**	86.04±0.21	78.47±0.31

In general, our method takes about 10 min to initialize the network and about 40 min to update. Moreover, we processed our HyperIID method on an NVIDIA Tesla V100-PCIE GPU.

C. Experimental Results on Synthetic Data

The decomposition results of different methods on synthetic data are shown in Fig. 10, and the comparison results about

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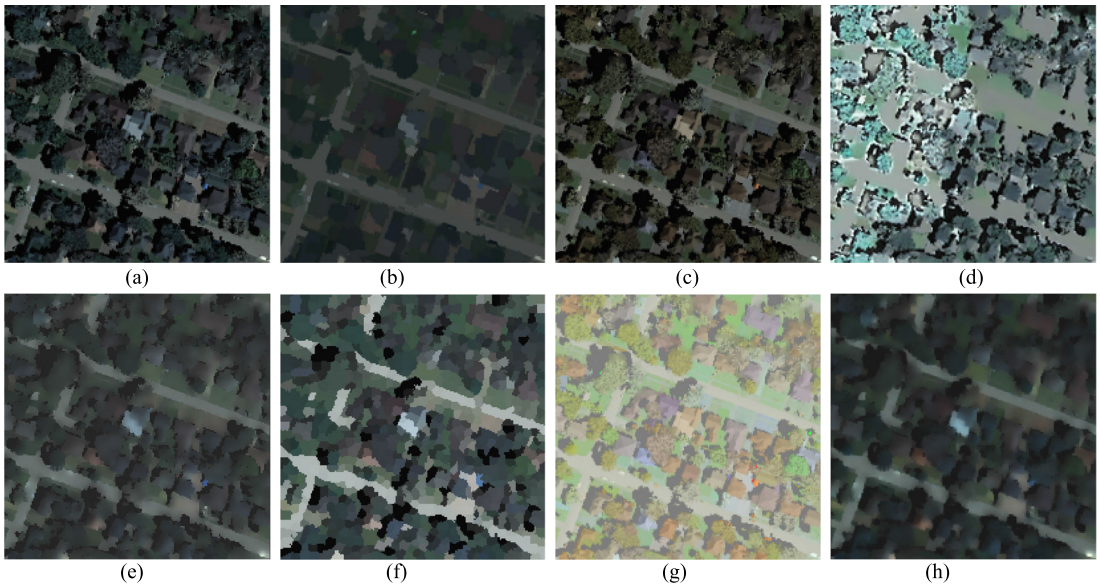


Fig. 10. Decomposition results comparison for synthetic data. (a) Original. (b) SIRFS. (c) UIDNet. (d) HIID. (e) SIID. (f) AFIID. (g) HyperIID-stage2. (h) Proposed.

TABLE VI
WCRV OF DIFFERENT WEIGHTS OF LOSS FUNCTION FOR THREE REAL DATASETS

λ_1/λ_2	Yellow River-area1	Yellow River-area2	Houston
0.2	38.4	34.8	7952.1
0.5	**37.4**	35.3	7898.4
1	38.2	34.9	8161.4
2	41.0	**34.2**	**7812**.1
5	41.0	34.5	7888.1
Ideal		↓0	

TABLE VII
MSE AND LMSE METRIC COMPARISON ON SYNTHETIC DATA

Methods	MSE			LMSE		
	reflectance	shading	average	reflectance	shading	average
SIRFS	0.0217	0.0729	0.0473	0.0215	0.0654	0.0434
UIDNet	0.0245	0.0626	0.0436	0.0194	0.0457	0.0326
HIID	**0.0063**	0.0160	0.0111	**0.0047**	0.0117	0.0082
SIID	0.0150	0.0391	0.0270	0.0136	0.0355	0.0246
AFIID	0.0065	0.0147	0.0106	0.0050	0.0098	0.0074
Proposed	0.0064	**0.0142**	**0.0103**	0.0048	**0.0094**	**0.0071**

MSE and LMSE metrics are shown in Table VII. We can see that the SIRFS method gets the worst decomposition results for quantitative MSE and LMSE metrics. Although SIRFS is a supervised method, it is designed for RGB images and is trained on small-scale laboratory objects, which are much different with large-scale remote sensing scenes, and this causes the poor decomposition performance of SIRFS method. For the unsupervised UIDNet method, it has a poor decomposition performance on the synthetic hyperspectral data whether for reflectance or shading components, since the network does not be developed with physical prior losses concerning remote sensing scenes to optimize the network parameters for each input hyperspectral data. While optimization-based HyperIID methods, including HIID and SIID, both have good decomposition performance for this synthetic hyperspectral data, since optimization-based methods are designed based on hyperspectral imaging principle of remote sensing scenes, leading to more reliable decomposition results for hyperspectral scenes. Specifically, the HIID method is superior as the reflectance and shading components decomposed by superpixel-based SIID method are affected by severe superpixel segmentation blocks. Since our proposed HyperIID network is driven by physical knowledge, compared with optimization-based methods, it gets a better optimized decomposition result by updating the parameters of network based on physical observation losses and prior knowledge-driven strategy. According to Table VII, the proposed HyperIID network gets best quantitative evaluation result on MSE and LMSE metrics.

V. CONCLUSION

In this article, a new unsupervised learning-based HyperIID network is proposed, which generates intrinsic reflectance from a single hyperspectral image with improved spectral homogeneity and classification performance. The HyperIID network comprises two encoder–decoder subnetworks, which aim to estimate the reflectance and shading components, respectively. The entire operation process is divided into two stages. In the first stage, a physical prior-driven model is implemented to initialize the parameters of the HyperIID network. In the second stage, HyperIID network integrates three reasonable loss functions based on physical observation to update the parameters. With the physical prior-driven system, the unsupervised learning-based HyperIID method demonstrates better intrinsic decomposition performance than other alternative methods, including learning-based and optimization-based methods, as evidenced by experimental results on classification tasks and reflectance recovery performance on three different hyperspectral datasets.

Future work should focus on developing learning-based systems that can address other issues related to hyperspectral intrinsic decomposition. Additionally, efforts should be made to enhance the computing efficiency of the estimation process.

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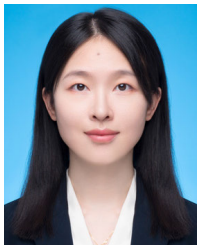
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